

Unsettling 'More Tech' Rhetorics in Educational Technology Research

A Case Study on Participatory Speculative Design in Rural Colombian Schools

Nicolas Ruiz

Centre for Research in Digital Education
University of Edinburgh
Edinburgh, United Kingdom
nicolas.ruiz@ed.ac.uk

Michael S Gallagher

Centre for Research in Digital Education
University of Edinburgh
Edinburgh, United Kingdom
michael.s.gallagher@ed.ac.uk

Cara Wilson

Centre for Research in Digital Education
University of Edinburgh
Edinburgh, United Kingdom
cara.wilson@ed.ac.uk

Jen Ross

Centre for Research in Digital Education
University of Edinburgh
Edinburgh, United Kingdom
jen.ross@ed.ac.uk

Abstract

Recognising that much HCI knowledge is built from the perspective of the Global North, this case study advances HCI efforts to understand how technologies, particularly educational technologies and innovation processes, can be understood from and in the Global South, through exploratory methods such as speculative design. We ran two workshops with twenty-four children aged five to eleven in rural schools in Colombia, exploring their perspectives on their education futures. We found that their visions of the future may support HCI practitioners to reconceptualise innovation and temporality in such contexts. We contribute one key principle, one provocation, and one practical learning which HCI researchers interested in decolonialising educational technologies may wish to consider in their own work.

CCS Concepts

• Human-centered computing; • HCI theory, concepts and models;

Keywords

Decoloniality, Pluriversality, Speculative Design, Campesinos, Children, Rurality

ACM Reference Format:

Nicolas Ruiz, Cara Wilson, Michael S Gallagher, and Jen Ross. 2025. Unsettling 'More Tech' Rhetorics in Educational Technology Research: A Case Study on Participatory Speculative Design in Rural Colombian Schools. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 9 pages. <https://doi.org/10.1145/3706599.3706699>

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CHI EA '25, Yokohama, Japan

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-1395-8/2025/04

<https://doi.org/10.1145/3706599.3706699>

1 Introduction

Educational technologies have been at the centre of discourses of educational innovation for some time [13], where they have been presented as a solution to pressing educational problems under the narrative of disruption and transformation [41]. Educational technologies are often positioned as fixes for 'broken' educational practices at the local level [17] or as an 'enhancement' of already existing practices [6]. Nonetheless, research has shown that educational technologies are far from what they promise: they fall short of their results [11, 13, 19] and widen educational inequalities [18, 49]. For instance, Facer and Selwyn [19], assessing the impact of 40 years of educational technologies, report that digital technologies do not transform education, nor improve learning, or fix inequalities; while any positive impacts are contextual and sociotechnical. Furthermore, the push for the incorporation of education technologies can have a harsher consequence in historically marginalised communities [12]. Among the many, a concrete example is the One Laptop per Child (OLPC) case in Paraguay, as researched by Morgan Ames [5]. Ames observed how the laptops were not used by the final users (rural students in Paraguay) for their designed purposes (by technological designers in the Global North), failing their charismatic promises at the expense of millions of dollars. For more shortcomings of educational technologies see: [30–32, 40, 41, 48, 51, 53]. In this context, it is crucial to re-think the purposes for which educational technologies are designed, mainly when designing technologies for marginalised communities.

This case study is born from the first author's doctoral research. He has undertaken ethnographies and speculative workshops in rural Colombian schools for six months. This research seeks to understand how educational technologies in rural Colombian schools can be purposed otherwise – beyond a "what-works" approach [43]. The what-works approach in educational research reduces research to "hard" evidence (e.g., randomised control trials) to reduce education's complexity and political nature. The what-works approach is also mirrored in technology design [8, 12]; techno-solutionism, driven through the quest for efficiency, disruption, and scalability, dismisses the ontopolitical nature of design [17]. Going beyond the "what-works" narrative allows us – researchers and practitioners –

to escape the traditional *modus operandi* of technologically-driven solutions: that same *modus operandi* that has resulted in shortcomings. Thus, this design process focuses not on the technologies' effectiveness, appropriation, or scalability but on understanding their purposes and their role in educational change from a local, bottom-up perspective. This case study presents the results of one of the five speculative design workshops done with 24 campesino students aged between 5 and 12. Campesinos are a heterogeneous cultural and political group fighting for political recognition in Colombia (see Section 1.2.1).

The workshops were undertaken within the more extensive framework of the first author's doctoral research, where he worked with 49 campesino students, four teachers, eight teachers, and 21 community members through speculative design workshops and ethnographies in three different locations. The speculative design workshops we present in this case study was structured as a scaffolding process, giving the young students multiple opportunities to envision and discuss what their school and educational systems might look like 100 years from now. The process involved creating technological prototypes through a magic machine [6, 9] approach and creating comics as speculative probes. As a case study, it puts forward several key reflections and insights useful for others exploring work around technologies, decoloniality, education, and rurality.

1.1 Related Work

This case study builds upon two lines of work within HCI. First, this work heeds the decolonial and postcolonial HCI call within HCI to re-think design methodologies, epistemologies, and technological embodiment from a critical sensitivity towards coloniality [3]. This case study is aligned with decolonial and participatory efforts to understand how indigenous and local populations and their knowledge are represented within HCI and how their future visions can guide technological development [4, 24]. Second, this work explores decolonising education and education technology with children through speculative and participatory methods; thus, this work aligns with design approaches that seek to empower children as primary agents of the design process [28].

1.1.1 Postcolonial, Decolonial, and Pluriversal HCI. In a recent systematic literature review, Das and Seeman [14] found that the HCI community has published 111 papers on "postcolonial computing" and "decolonial computing". These two complementary concepts propose re-examining, through a critical lens, traditional narratives of development that guide technological design and deployment [4, 27]. From recognising that most HCI projects are developed in the Global North [34], postcolonial computing brings awareness that technologies migrate from the Global North towards the Global South. This technological migration tends to bypass target 'users' sociocultural realities and might fail to recognise the entanglements of capital, technological design, and development discourses [27]. In the case of educational technologies, this technological limitation tends to be the norm.

Decolonial computing aims to transform HCI into a world of many worlds ("Un mundo donde quepan muchos mundos" [16] – "a world in which many worlds fit"). In Alvarado et al.'s [4] decolonial manifesto, five analytical pathways are proposed to achieve this

goal. The five pathways invite the HCI community to be mindful of their paradigms, methods, participation issues, teleology, and issues of power. Regarding methods and participation, the manifesto provides reflection prompts about diversifying our knowledge production methodologies to incorporate voices, ontologies, and epistemologies from the peripheries. This epistemological path is open to exploring, from a non-extractivist position, how technologies can be designed otherwise while being mindful of who designs and why. A recent example of decolonial HCI research is Guerrero-Millán et al.'s work on data and information services [24]. Guerrero-Millán et al.'s work sought to explore data perception and usage and visionary technological projects with the Masewal indigenous community in Mexico. They worked through participatory and speculative methods to explore how the Masewal cosmivision could be aligned with technological development.

To expand decolonial approaches in HCI, we can approach two related disciplines that can enhance the conversation within HCI: design justice [12] and minimal computing [35, 42]. Design justice is a design framework that seeks to respect the knowledge, values, and practices of the communities that will be the "end users" of technological products. This framework aims to address power imbalances and amplify the voices of marginalised groups in tackling complex challenges: "We centre the voices of those who are directly impacted by the outcomes of the design process" [12]. Minimal computing is a "conceptual provocation" [22] that promotes the use of only the minimum needed technologies for developing digital work in resource-constrained environments [42, 46]. Minimal computing resists the notion of "innovation" defined by newness, scale, or scope [42]. As an ethos, minimal computing empowers users to develop technological proficiency and gain technological autonomy [35].

To conclude this section, we want to bring attention to Escobar's seminal work on pluriversal design [17]. Escobar traces how design has had a paradigmatic turn towards participation, sustainability, social justice, decoloniality, and pluriversality [17]. This hopeful turn recognises that design is ontopolitical and that effort is needed on the many design fronts to overcome the current climate and social injustice challenges. This paradigm change goes from "elimination design" to "sustainable design" [20]; we are transitioning from hegemonic design practices that eliminate all vernacular, indigenous, and autonomous ways of being – while eliminating possible hopeful futures – towards design practices for a world where many worlds fit.

1.1.2 Children as Protagonists: Participation and Speculation. The Child Computer Interaction (CCI) community has a history of promoting children as active participants in design. Nonetheless, how children participate in said processes remains a topic of discussion. In worst-case scenarios, children participate tokenistically [25], but it is most common to find that children are primarily included in ideation processes but not as empowered decision-makers [33]. To expand children's role in participatory design, Iversen et al. [28] propose considering them as protagonists of the design process. This paradigmatic change in child research and design [37] seeks to design with them (and in the best case, they design by themselves) contrary to the common locus of design: us (the adults) designing for them.

Following this politics of participation, this case study delves into participatory speculative design with children. First, it draws from previous research on magic machines. The magic machine is a workshop methodology that scaffolds on playful traditions, embodied design, cultural probes, speculative design, and critical making [6]. Through imagination, children are invited to answer a design challenge that stretches possibilities. Andersen [6] proposes the following structure for magic machine workshops: introduction, prompt, making, description, group discussion, and documentation.

Second, this case study also builds on using comics as speculative cultural probes. As a speculative design method, comic-making can critically explore alternatives to the dominant frameworks of future-making and technology-making. For those involved in community-based participatory design, this approach to exploring alternative futures enables individuals to re-think their surroundings and the systems within them, using familiar materials. Exploring alternative futures from a speculative stance avoids predictions and anticipatory regimes [44] while being critical of technological progress and determinism. More broadly, speculative methods serve as decolonial fugue points.

This critical approach to technology is crucial when collaborating with marginalised rural communities. First, it allows a decolonial stance towards the future by disassociating "progress" from "more technology" (a standard coupling found in development projects). This disassociation is decolonial because it opens a space for the participants' autonomy by not imposing dominant visions of the future. Second, it opens speculative space for the unexpected. Thus, this research distances itself from asking "what works" to embrace uncertainty and complexity [43]. Finally, it allows us to question who imagines the future, recognising that historically marginalised communities are not commonly included in the narratives of future and technological design [12].

1.2 Colombian Context

1.2.1 Campesinos. Campesinos in Colombia are a heterogeneous group characterised by their cultural, political, and economic practices [19]. Among the many characteristics of this group, culturally, they build solid communal relations with their extended family and neighbours. These relationships configure how campesinos work the land and their care networks. Regarding labour, campesinos rely on family work for substance agricultural practices, evidenced by child labour [9]. Children tend to work the land from a young age. For instance, the children observed in this ethnographic work would go home after school (which ended at 12:30 pm) to do their homework and work on the crops.

Campesinos have been building an agenda around land redistribution and political recognition. Land redistribution has been a locus of injustice for campesinos since the birth of the Colombian republic [30]. Historically, criollos (Spaniards born in "the new world") have taken advantage of their position in the social system to appropriate large portions of land. In the 20th century, armed conflict, campesino dispossession and forced migration (mostly to live in cities in poverty) have intensified the land problem.

Regarding political recognition, campesinos have been fighting to be recognised as political beings, though they did not achieve this goal when the Colombian Constitution of 1991 was promulgated.

On the other hand, indigenous and afrocolombian communities achieved political recognition in this Constitution. Political recognition allows marginalised communities autonomy, budget, and public policies for their needs. From an educational perspective, Indigenous and afrocolombian communities have achieved educational autonomy – within certain limits imposed by the system – through the ethnoeducation law and public policies. Educational autonomy is not the case for campesino communities, as they must adhere to urban educational policies and frameworks.

1.2.2 Colombian Rural Education. Regarding education, campesinos fall under the "rural education" label within the Colombian educational system. The Ministry of Education uses the term "rurality" to refer to a diverse group of non-urban residents, including marginalised communities in Colombia, such as Afro-Colombians, indigenous peoples, and campesinos. Quantitative and qualitative indicators indicate that rural schools have the most vulnerabilities in the Colombian public educational system. For example, rural students tend to leave the educational system after 5.7 years of education [39]. The educational inequities that campesino students suffer stem from the detrimental effects of the armed conflict, demographical diminishing, decontextualisation of the curriculum, and the digital divide, among others. To overcome these issues, Colombia has had a history of public policy that has had minor effects in changing the situation of rural education for campesinos [39]. For instance, the "Special Plan for Rural Education" (SPRE) is one of these efforts. The SPRE was born from the peace treaty between the Colombian Government and the FARC guerrilla in 2016.

Regarding educational technologies and public policy, CONPES 3988 is the latest policy document that governs educational technologies within the Colombian educational system. CONPES 3988 is a Colombian public policy document that governs educational technologies. CONPES 3988 was built upon two other policy documents: CONPES 3063 and CONPES 3975. CONPES 3063 governed the provision of laptops to public schools, and CONPES 3975 integrated teacher professional development focused on the use of educational technologies – the provided laptops. Both CONPES 3988 and the SPRE seek to incorporate ICT into education through "educational innovation" to improve educational quality. Educational innovation is understood here as a process in which educational practices, with the mediation of ICT, are reconfigured to address a contextual need, expectation, or problem [39].

2 Statement of Positionality, Ethics, and Reciprocity

The first author is a male born and raised in Colombia and has heritage connections to Tunja, the site of this work. He has also worked with the participant community since August 2020 in several educational development projects through a non-profit organisation. He was raised in an upper-middle class household in Bogotá, distant from rural contexts and practices. Nonetheless, he has generated connections with rural territories and their people throughout his life. The second author is a Scottish woman researcher and teacher engaged in participatory design, predominantly with people who are historically overlooked in design processes. She is PhD supervisor to the first author. The third author is male, a citizen of both the

Table 1: Demographic and group information of the participants

Participant	Age	Group
P1	5	Group 2
P2	5	Group 2
P3	5	Group 2
P4	5	Group 2
P5	5	Group 2
P6	6	Group 2
P7	6	Group 1
P8	6	Group 1
P9	6	Group 1
P10	6	Group 1
P11	6	Group 1
P12	6	Group 1
P13	7	Group 2
P14	7	Group 2
P15	7	Group 2
P16	7	Group 2
P17	8	Group 2
P18	8	Group 2
P19	9	Group 2
P20	9	Group 2
P21	9	Group 2
P22	10	Group 1
P23	11	Group 1
P24	11	Group 1

US and the UK, and works extensively with refugee populations largely in Sub-Saharan Africa. The fourth author is a dual citizen of the UK and Canada, working on speculative methods mostly in higher education and museum contexts in the UK and Europe. All of the authors recognise the power asymmetries of their socioeconomic backgrounds and work to balance said asymmetries in this research through participatory research methods. The authors recognise the right to autonomy and justice of the campesinos of Colombia. This case study was approved by the University of Edinburgh ethics committee in the context of the author’s doctoral research.

3 Case Study

3.1 Location and Participants

This case study takes place in one of the veredas (rural ‘neighbourhoods’) of Tunja. Tunja is the capital of the Boyacá department. It has 172,000 inhabitants, of which 8,000 are in rural areas. The history of Tunja remotes to the Muisca, the most numerous pre-Colombian indigenous community in Colombia [38]. Tunja’s economy is based on its universities, service industry, and agriculture.

The participants of this case study are all self-identifying campesino students from a rural public school. Twenty-four participants, children between 5 and 12, participated in the workshop (see Table 1). The participants were divided into two groups by school grade; ‘group 1’ consisted of first grade, fourth grade, and fifth grade, and ‘group 2’ consisted of transition grade, second grade, and third grade. The complete list of participants can be found in

Appendix A. Each group had a workshop (hence, two workshops happened). The workshops were conducted in six sessions, each lasting two hours.

In terms of digital literacy, the participants had computational thinking classes for four months before the workshops. In these classes, they learned about digital devices and basic algorithms. They had access to a programmable robot called ROVERSA [45], designed by the Global Center for Equitable Computer Science Education, and laptops provided by the Colombian Government. As observed in ethnographic work, older students (10 to 12 years old) were accustomed to using laptops to navigate the Internet by the school. Also, all students discussed that they used their parents’ mobile phones, but none of them owned one. Thus, participants were familiar with digital technologies and, arguably, they had basic digital literacy skills.

3.2 The Workshops

The speculative design workshops used a magic machine approach [6] and speculative comics [10]. As the central trope of the workshop, students were invited to imagine a technology that they would like to be present in their school in 100 years. The workshop involved an introductory phase where students were presented the concepts of past, present, and future. These concepts were presented through the exposition of old photos of schools and students and through dialogue to clarify possible misconceptions about these concepts. The photos were helpful as they presented a nuanced approach to change, a crucial caveat when undergoing speculative

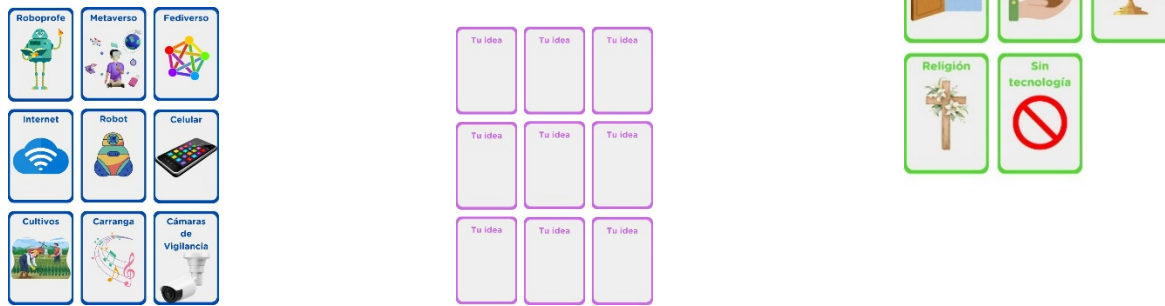


Figure 1: Conversational cards

workshops with young children as they might fall into fantastical imagination (i.e., the aliens will invade the school in 100 years).

The second phase of the workshop involved providing cards as conversational prompts (see Figure 1). The cards had images of current and possible technologies, objects of their everyday life, and concepts relevant to their culture. Other “empty” cards allow students to paint their ideas. The cards were used in group discussions to settle the focus on technological approximations to the coming speculative efforts. Notably, students were asked: “Is this ‘element’ going to be present in 100 years?” and “Is this ‘element’ important?”. These two questions “warmed up” the workshops so students could work on their own discussed ideas in the following phases.

The third phase of the workshop involved a magic machine approach. Students were prompted to work in groups or by themselves to prototype an object they would enjoy seeing in their schools in 100 years. The prototyping process began with a prompt: “What technology would you like to see in this school in 100 years?”. To clarify this prompt, the first author would ask them to imagine the future students by imagining they would be their grandchildren. This prompt helped the youngest students imagine who they were designing for, as family bonds are central to campesino communities. Students were asked to brainstorm with their groups possible technologies they would like to see in 100 years. Some groups would use some of the ideas provided in the conversational cards. Afterwards, they were invited to pick one of the ideas and discuss it with the first author.

Regarding power dynamics in the design process, the first author had two roles. First, he was the “guardian” of the materials to make the prototypes. As a “guardian” he provided the needed materials to the groups who had a clear idea about what they would like to make. By “clear idea”, the criteria is that they had achieved consensus as a group upon a prototype. The second role involved asking questions to better understand the purpose and affordances of the technology. For example, in the case of the ToboRegador, P23 mentioned that the slide would have a slide to water the crops, to which author 1 asked to clarify which crops were the ones she mentioned. Participants’ answers were recorded in a field log.

To make the prototypes, students had access to different types of materials. Expressly, students were provided recycled materials (paper roll cardboard, egg boxes, cardboard boxes, plastic containers), plastic building blocks, sparkling foam paper, clay, paint, yarn, glue, scissors, tape, and resources they could get from their environment (plants and rocks). This phase finished with a group presentation of each of the prototypes and with the documentation of the magic machines.

Finally, in the last phase, students were invited to create a speculative comic to explore how that object was used in their schools on a typical day. Students were introduced to comics through group reading of age-appropriate comics. In group sessions, we discussed the main elements of the stories: main character, plot, and setting. Afterwards, students were given graphical organisers to help them design their characters, plot, and setting. When students were finished using the graphic organisers, they developed their comics on provided “comic pages”. After completing the comics, they presented their stories to the researcher and the other students.

3.3 Results: Prototypes and Comics

As a result of the workshops, six prototypes and four comics were created. Although the youngest students (ages five and six) tried to create comics, the task exceeded their current skills and did not motivate them as much as the prototyping.

3.3.1 The “ToboRegador”: a waterslide over the garden. A pair of students designed a waterslide that allows students to waterslide while watering the crops beneath the slide. The waterslide is a structure composed of a staircase, a slide, a pool where students land, a hose to allow water to flow from the pool to the slide, and a ‘robotic’ system that waters the crops when nobody uses the slide. The waterslide is intended to be a fun but useful technology. The designers emphasised the importance of having crops at their schools; they would like to grow potatoes, peas, broad beans, and corn. Currently, there are no crops in the school, and the curriculum does not work with any vernacular agricultural knowledge. Paradoxically, student alimentation in this school depends on a public program that sends processed food.

The comic created about the ToboRegador tells how the waterslide breaks and then is fixed by a team of students. The story

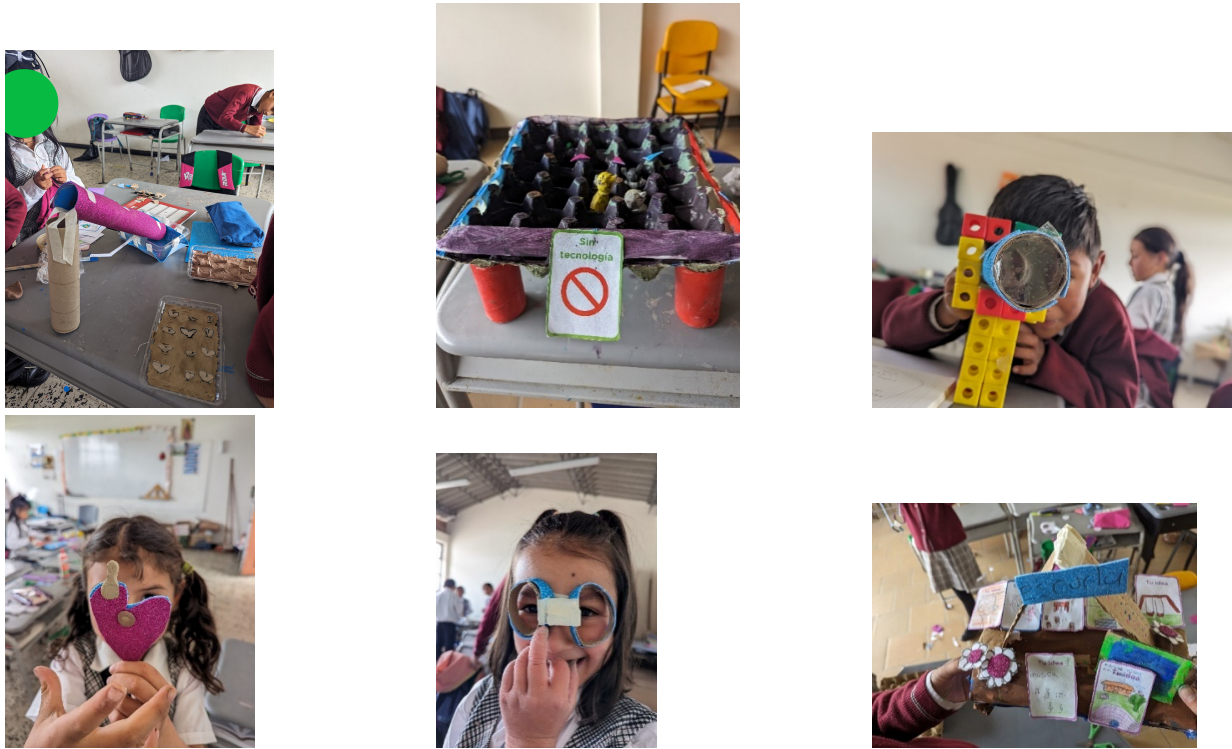


Figure 2: The six prototypes. From left to right, beginning in the upper row: the ToboRegador, the animal museum, the Really Big Telescope, the Heart-shaped Phone, the Internet Glasses, and the School Model

begins in *media res*, as three students notice the ToboRegador leaking. After fixing the waterslide, the three students proceed with their morning. First, they visit the hen and the chickens in the school. Afterwards, they visit the horse and the crops. It is essential to mention that the school has no animals or crops at the moment of the workshop.

3.3.2 The Animal Museum. Three students imagined what it would be like to have animals at school. After trying several ideas, they arrived at an animal museum. The museum would hold wax wild animals so students can learn more about them. The students did not allow mobile phones or other technologies inside the museum. On the other hand, the museum would have a small prayer room inside where students can pray so “*nothing bad happens*”. Initially, they had thought about including security cameras to protect the museum but desisted the idea of having zero digital technologies inside the museum. Instead, they believe that the prayer room and the cross will be enough protection for the museum.

The comic related to this prototype tells how the museum was built. It is a short story about how “Lucas”, the Mayor of Tunja, builds the museum as a gift for the students. Afterwards, the indigenous communities use clay to build the animals hosted in the museum.

3.3.3 The Really Big Telescope. A group of three students designed a building supporting a big telescope. The building mainly comprises stairs and a telescope at the end of the staircase. The telescope

is designed to be big and powerful enough to see the stars. However, it is also designed to allow students to see far away inside the vereda, know if there are “*any accidents*”, to watch the cyclist go uphill, and “*do animal looking*”.

Two short stories were born out of this prototyping exercise. One comic tells the story of a student who uses the telescope to watch the cyclists cycle in a far-away mountain. The other story is not a comic but told through the graphic organiser. In this story, a student uses a telescope to see a fire burning some trees in the distance. The student who wrote the story is worried about possible wildfires, so the telescope would allow students to prepare for them and take care of their environment.

3.3.4 The Heart-Shaped Phone. Four students designed a heart-shaped mobile phone that all students would have in 100 years. The phones would allow students to listen to music, take pictures of animals, share those pictures, and watch videos. The phone would have two main differences from current phones: first, they would have a heart-shape and, second, the phones would let them know if somebody is running rumors about them. No comics were done to complement the prototypes.

3.3.5 The Internet Glasses. Three students designed glasses that have three purposes. First, the glasses are designed to magnify small animals, such as bugs. Seeing small animals would allow them to “*learn what they are and what they are not*”. Second, they are similarly connected to the Internet to the metaverse, meaning

the wearer can access a virtual reality. Third, through their internet connection, the glasses would be able to analyse bugs in real-time to provide information to the wearer. One of the students was keen on knowing if the analysed bugs were dangerous.

The comic created after the glasses was titled "Internet Glasses" and depicts a girl accessing the metaverse with the help of her glasses. In the metaverse, she first sees a door. After opening the door, she sees her glasses inside the room. Afterwards, she sees another male student beside a TV. Then she sees, in order of appearance, a portable book, a clock, a sleepy pencil case, a travelling cap, and a school where students learn about technology.

3.3.6 A miniature model of the classroom. The last group, of three students, designed a model of their classroom in 100 years. The model included several details of the classroom. For instance, they thought of including a tree and a flower garden inside the classroom and having crops and a football field outside the classroom. The classroom will have a chalkboard (right now, it has a whiteboard) and laptops with the Internet. Interestingly, the classroom they designed does not differ as much to their current classroom. Their designed classroom does not differ as much from their current one. This model does not have a comic.

4 Discussion

Undertaking participatory speculative design workshops with young campesino students in Tunja provided insights into the challenges posed by postcolonial and decolonial HCI efforts. In this section, we present a guiding principle, a provocation, and practical learning for HCI's future research with campesino communities and other communities from the Majority of the World.

4.1 Principle: Innovation is more-than-digital

The first principle involves having an open mindset about what innovation means. Traditionally, innovation is understood as developing cutting-edge technologies and advancing the frontier of what technologies can do [42]. This understanding of innovation is crystallised in sociotechnical imaginaries [29] as the Fourth Industrial Revolution [7]. In counterpoint to these discourses and imaginaries, we invite the HCI community to embrace a more-than-digital mindset when designing with/for marginalised communities.

We chose the phrase "more-than-digital" because these communities are also part of the postdigital [44] (Internet, mobile phones, and computers are part of their social assemblages), meaning that we do not want to negate digital technologies as materials of innovation. Nonetheless, our invitation is not to prioritise digital technologies and digital proxy values: efficiency and newness (among others). Instead, we suggest prioritising more-than-digital values to frame innovation and the future. These values are not fixed; they stem from the specificity of each local community that engages in innovation processes - thus, it is not a fixed list of preconceived 'desirables'. Instead, these more-than-digital values will emerge when encountering specific contexts, territories, and communities.

In our case study, four of the magic machines created did not prioritise digital forms of technology: the ToboRegador, the animal museum, the school model, and the telescope. One of the prototypes - the museum - explicitly prohibited using digital technologies. At the centre of these creations, the students have other priorities

when thinking about the future. The animal museum prototype demonstrates that their connection to nature, animals, and territory is more critical than any cutting-edge technology. The telescope also demonstrates their connection towards the physical world and their community; it allows students to see far-away and to protect their surroundings. To sum up, we highlight that several more-than-digital values emerged from the workshops: nature, fun, and food sovereignty are some of them.

In the Colombian case, this principle can also inform educational policy design. As discussed earlier, Colombian educational policies, namely CONPES 3988, that govern educational technologies mostly align with the hegemonic vision of innovation and techno solutionism. These policies could benefit from this principle by incorporating other imaginaries and values into educational technology programs. We ask: how would it look like to have more-than-digital values guiding educational technology policies? What would policy look like if it builds upon imaginaries beyond northern speculations - as the 4R [7]? Answering these questions involves detaching innovation from the "more tech" rhetoric.

Finally, the more-than-digital mindset aligns with the pluriversal call of decolonial and postcolonial HCI. Accepting different teleological values in technological design and innovation enables the emergence of projects that do not have to align with the single-world hegemonic narrative. Autonomy, diversity, and vernacular design will have a space to grow.

4.2 Provocation: We can future towards the "authentic present".

This provocation involves being sensitive about what we consider the future to be when engaging in speculative and futuring methods. Concretely, this provocation invites the HCI to reconsider a central theme in speculative methods: temporality. This provocation suggests using non-linear temporalities to imagine the future, such as the spiral temporality used by Indigenous communities in Colombia [49]. Approaching temporality in non-linear ways allows HCI researchers and practitioners to consider the future as other presents and pasts, opening teleological distance from a single linear vision of development (usually, Western development discourses). Assuming that development is not one (but many) will open space for pluriversal practices within HCI.

In our case study, the designed prototypes and comics highlighted how campesino students are alienated from their context and prefer a future akin to their "authentic present". Alienation happens when the curriculum, didactics, assessment and technologies are directed by the urban/hegemonic development logics - as evidenced in these speculative probes. For instance, the ToboRegador prototype and comic discuss how students value their present agricultural and husbandry traditions. Students recognise the importance of preserving their knowledge and practices; the designed technologies embody such concerns. In this manner, instead of envisioning a future where cutting-edge technologies dictate innovation, campesino students seek a future that aligns with their "authentic present". By "authentic present", we refer to their campesino lifeworld: working the land, taking care of animals, relating with nature, and their communal relations - among the many social, territorial, cultural, and productive characteristics of their everyday

life. In this way, the speculative probes suggest a future where we return to the present.

Returning the present (in this case, the authentic present) in the future opens the possibility of considering temporalities beyond the linear in speculative methods - common in postcolonial and decolonial research within HCI. Instead of a line, we can depict the future as a spiral, such as the ones used by several indigenous communities in Colombia [26].

4.3 Practical Learning: Community building as a method

Before these workshops happened, the first author engaged in community-building processes with this community. Said author has worked with this school, the parents, teachers, and school leaders since 2020, when he helped the school navigate school closure during the pandemic through fundraising for laptops and Internet and teacher professional development. More recently, in the first semester of 2024, this author spent two months doing ethnographical work and connecting with the community. Connection with the community happened through play, sharing meals, and learning from one another. Thus, sustained connection and dialogue have generated a relationship between the researcher and the community based on trust and understanding.

We learned that community building can be a method when working with any community, as design interactions do not happen in a vacuum. We recognise that design interactions happen in specific contexts and are done by humans. Humans need time and space to know and trust each other, conditions allowing collaboration and design towards decolonial futures.

4.4 Challenges and Limitations

This case study has contextual and methodological limitations. Contextually, this study does not seek to universalise its findings towards the whole Colombian campesino community. Instead, it recognises that these findings speak about the visions of the future of this particular group of campesino children.

Methodologically, we learned that speculative comics are not an appropriate methodology for children aged 5 to 7. While the magic machine was compelling to all children, the youngest children had trouble keeping engaged with speculative comics. We had prepared scaffolding to allow the youngest children to communicate themselves through drawings and graphic organisers, but it was not enough for most of the youngest children.

5 Closing Reflections

It is vital to foresee how technology pushes (or undermines) neo-colonialism into rural schools through discourses of development and innovation, and it is also essential to recognise the decolonial visions of the future of marginalised communities. In this case study, we propose a principle, a provocation, and a practical lesson to help us create a pluriversal HCI community—first, the principle: more-than-digital technological futures can be innovative. Second, we provoke by stating that we can future towards the “authentic present”. The path to the future can be guided by recognising and honouring present cosmovisions and ways of being – instead of taking these communities towards new and “disruptive” futures.

Third, we learned that community building can be considered a methodology in HCI decolonial research.

Future research on this strand can develop in two manners. First, decolonial and postcolonial research within HCI can be enriched by considering the minorities within the Global South – such as Colombian campesinos. As a totalising label, the “Global South” tends to render invisible the diversity of power relations within southern polities beyond the Global North/Global South dichotomy. In such a sense, another geopolitical labelling - “the Majority of the World” [2] - can be more appropriate to nuance coloniality within southern polities.

Second, building on this case study, further research can focus on community building with communities of the Majority of the World. Community building in this case study began as a partnership relation [15] instead of a dominant or extractivist one. The challenge for us researchers, is to bridge human bonds beyond the self-interest of publishing and data extraction; doing so renders again neocolonial relations. We invite CHI researchers to befriend and care for their communities and build long-lasting bonds.

References

- [1] Muhammad Sadi Adamu. 2023. No More “Solutionism” or “Saviourism” in Turing African HCI: A Manyfesto. *ACM Trans. Comput.-Hum. Interact.* 30, 2: 21:1–21:42. <https://doi.org/10.1145/3571811>
- [2] Shahidul Alam. 2008. Majority World: Challenging the West’s Rhetoric of Democracy. *Amerasia Journal* 34, 1: 88–98. <https://doi.org/10.17953/amer.34.1.13176027k4q614v5>
- [3] Syed Mustafa Ali. 2016. A brief introduction to decolonial computing. *XRDS* 22, 4: 16–21. <https://doi.org/10.1145/2930886>
- [4] Adriana Alvarado Garcia, View Profile, Juan F. Maestre, View Profile, Manuhua Barcham, View Profile, Marilyn Iriarte, View Profile, Marisol Wong-Villacres, View Profile, Oscar A Lemus, View Profile, Palak Dudani, View Profile, Pedro Reynolds-Cuellar, View Profile, Ruotong Wang, View Profile, Teresa Cerratto Pargman, and View Profile. 2021. Decolonial Pathways: Our Manifesto for a Decolonizing Agenda in HCI Research and Design. *Extended Abstracts of the 2021 CHI Conference on Human Factors in Computing Systems*: 1–9. <https://doi.org/10.1145/3411763.3450365>
- [5] Morgan G. Ames. 2019. *The Charisma Machine: The Life, Death, and Legacy of One Laptop per Child*. MIT Press.
- [6] Kristina Andersen and Ron Wakkary. 2019. The Magic Machine Workshops: Making Personal Design Knowledge. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*, 1–13. <https://doi.org/10.1145/3290605.3300342>
- [7] James Avis. 2018. Sociotechnical imaginary of the fourth industrial revolution and its implications for vocational education and training: a literature review. *Journal of Vocational Education & Training* 70, 3: 337–363. <https://doi.org/10.1080/13636820.2018.1498907>
- [8] Sian Bayne. 2015. What’s the matter with ‘technology-enhanced learning’? *Learning, Media and Technology* 40, 1: 5–20. <https://doi.org/10.1080/17439884.2014.915851>
- [9] Mark Blythe, Kristina Andersen, Rachel Clarke, and Peter Wright. 2016. Anti-Solutionist Strategies: Seriously Silly Design Fiction. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems*, 4968–4978. <https://doi.org/10.1145/2858036.2858482>
- [10] Kirsten Bray and Christina Harrington. 2021. Speculative Blackness: Considering Afrofuturism in the Creation of Inclusive Speculative Design Probes. In *Proceedings of the 2021 ACM Designing Interactive Systems Conference (DIS’21)*, 1793–1806. <https://doi.org/10.1145/3461778.3462002>
- [11] Ruy Cervantes, Mark Warschauer, Bonnie Nardi, and Nithya Sambasivan. 2011. Infrastructures for low-cost laptop use in Mexican schools. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI’11)*, 945–954. <https://doi.org/10.1145/1978942.1979082>
- [12] Sasha Costanza-Chock. 2020. *Design Justice: Community-Led Practices to Build the Worlds We Need*. 358.
- [13] Larry Cuban and Petar Jandrić. 2015. The dubious promise of educational technologies: Historical patterns and future challenges. *E-Learning and Digital Media* 12, 3–4: 425–439. <https://doi.org/10.1177/2042753015579978>
- [14] Dipto Das and Bryan Semaan. 2022. *Decolonial and Postcolonial Computing Research: A Scientometric Exploration*. In *Companion Publication of the 2022 Conference on Computer Supported Cooperative Work and Social Computing*

- (CSCW'22 Companion), 168–174. <https://doi.org/10.1145/3500868.3559468>
- [15] Riane Eisler. 2012. *Sacred Pleasure: Sex, Myth, and the Politics of the Body*. Harper Collins.
- [16] Zapatista de Liberación Nacional Ejercito. 2020. CUARTA DECLARACIÓN DE LA SELVA LACANDONA. Enlace Zapatista. Retrieved October 11, 2024 from <https://enlacezapatista.ezln.org.mx/1996/01/01/cuarta-declaracion-de-la-selva-lacandona/>
- [17] Arturo Escobar. 2018. *Designs for the pluriverse: radical interdependence, autonomy, and the making of worlds*. Duke University Press, Durham.
- [18] Rebecca Eynon and Lars-Erik Malmberg. 2021. Lifelong learning and the Internet: Who benefits most from learning online? *British Journal of Educational Technology* 52, 2: 569–583. <https://doi.org/10.1111/bjet.13041>
- [19] Keri Facer and Neil Selwyn. 2021. Digital Technology and the Futures of Education: Towards 'Non-Stupid' Optimism. Retrieved December 21, 2022 from <https://research.monash.edu/en/publications/digital-technology-and-the-futures-of-education-towards-non-stupid>
- [20] Tony Fry. 2010. *Design as politics*. Berg.
- [21] Michael Gallagher. 2019. Educational Unsustainability in Sub-Saharan Africa: In Search of Counter-Narratives to Policy Pressures and Exponential Tech Growth. *Visions for Sustainability: No 12* (2019) with Special Section "Slow Tech." <https://doi.org/10.13135/2384-8677/4040>
- [22] Alex Gil and Elika Ortega. 2016. Global outlooks in digital humanities: multilingual practices and minimal computing. In *Doing Digital Humanities*. Routledge.
- [23] Marta Saade Granados. 2020. Conceptualización del campesinado en Colombia: documento técnico para su definición, caracterización y medición. ICANH.
- [24] Carlos Guerrero Millan, Bettina Nissen, and Larissa Pschetz. 2024. Cosmovision Of Data: An Indigenous Approach to Technologies for Self-Determination. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*, 1–13. <https://doi.org/10.1145/3613904.3642598>
- [25] Roger A Hart. 2008. Stepping back from 'The ladder': Reflections on a model of participatory work with children. In *Participation and learning: Perspectives on education and the environment, health and sustainability*. Springer, 19–31.
- [26] Edgar Gabriel Suárez Higuera. 2020. Una Reinterpretación Pictográfica De La Espiral En Algunas Culturas Indígenas De Colombia. *Lingüística y Literatura*, 77: 433–469.
- [27] Lilly Irani, Janet Vertesi, Paul Dourish, Kavita Philip, and Rebecca E. Grinter. 2010. Postcolonial computing: a lens on design and development. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI' 10)*, 1311–1320. <https://doi.org/10.1145/1753326.1753522>
- [28] Ole Sejer Iversen, Rachel Charlotte Smith, and Christian Dindler. 2017. Child as Protagonist: Expanding the Role of Children in Participatory Design. In *Proceedings of the 2017 Conference on Interaction Design and Children*, 27–37. <https://doi.org/10.1145/3078072.3079725>
- [29] Sheila Jasanoff and Sang-Hyun Kim. 2015. *Dreamscapes of Modernity: Sociotechnical Imaginaries and the Fabrication of Power*. University of Chicago Press.
- [30] Jeremy Knox, Ben Williamson, and Sian Bayne. 2020. Machine behaviourism: future visions of 'learnification' and 'datafication' across humans and digital technologies. *Learning, Media and Technology* 45, 1: 31–45. <https://doi.org/10.1080/17439884.2019.1623251>
- [31] Janja Komljenovic. 2021. The rise of education rentiers: digital platforms, digital data and rents. *Learning, Media and Technology* 46, 3: 320–332. <https://doi.org/10.1080/17439884.2021.1891422>
- [32] Janja Komljenovic and Susan L. Robertson. 2016. The dynamics of 'market-making' in higher education. *Journal of Education Policy* 31, 5: 622–636. <https://doi.org/10.1080/02680939.2016.1157732>
- [33] Monica Landoni, Elisa Rubegni, Emma Nicol, and Janet Read. 2016. How Many Roles Can Children Play? In *Proceedings of the The 15th International Conference on Interaction Design and Children (IDC' 16)*, 720–725. <https://doi.org/10.1145/2930674.2932222>
- [34] Sebastian Linxen, Christian Sturm, Florian Brühlmann, Vincent Cassau, Klaus Opwis, and Katharina Reinecke. 2021. How WEIRD is CHI? In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems (CHI' 21)*, 1–14. <https://doi.org/10.1145/3411764.3445488>
- [35] Andrew Lison. 2022. *Minimal computing*.
- [36] Absalón Machado Cartagena. 2021. *La ruralidad que viene y lo urbano: un despertar de la conciencia*. Siglo del Hombre Editores.
- [37] Jan Mason and Susan Danby. 2011. Children as Experts in Their Lives: Child Inclusive Research. *Child Indicators Research* 4, 2: 185–189. <https://doi.org/10.1007/s12187-011-9108-4>
- [38] Jorge Orlando Melo. 2017. *Historia mínima de Colombia*. El Colegio de México AC.
- [39] Ministerio de Educación. 2018. *Plan especial de Educación Rural*.
- [40] Curtis Riep. What do we really know about Bridge International Academies?
- [41] Curtis B. Riep. 2017. Making markets for low-cost schooling: the devices and investments behind Bridge International Academies. *Globalisation, Societies and Education* 15, 3: 352–366. <https://doi.org/10.1080/14767724.2017.1330139>
- [42] Roopika Risam and Alex Gil. 2022. Introduction: The Questions of Minimal Computing. *Digital Humanities Quarterly* Volume 16.
- [43] Jen Ross. 2017. Speculative method in digital education research. *Learning, Media and Technology* 42, 2: 214–229. <https://doi.org/10.1080/17439884.2016.1160927>
- [44] Jen Ross. 2023. Postdigital Speculation. In *Encyclopedia of Postdigital Science and Education*, Petar Jandrić (ed.). Springer Nature Switzerland, Cham, 1–5. https://doi.org/10.1007/978-3-031-35469-4_19-1
- [45] Roversa Robotics. Home. Roversa Robotics. Retrieved January 5, 2025 from <https://www.roversa.com/home.html>
- [46] Nicolás Ruiz, Javier Tejera, Osmany Sabalza, Steffany Soache, and Gisella Jassir. Rebusque and minimal computing in rural Colombia: LibreEscuela, an OER co-creation project. *Learning, Media and Technology* 0, 0: 1–13. <https://doi.org/10.1080/17439884.2024.2387227>
- [47] Natalia Sánchez Corrales, Álvaro Hernández Bello, and Catalina López Gómez. 2019. La escuelas del río: una lectura del Plan Especial de Educación Rural. *Revista Universidad de La Salle* 1, 79: 111–130. <https://doi.org/10.19052/ruls.vol1.iss79.6>
- [48] Neil Selwyn. 2023. Lessons to Be Learnt? Education, Techno-solutionism, and Sustainable Development. In *Technology and Sustainable Development*. Routledge.
- [49] Gory Soto Mella. 2023. En lineal y en espiral: una mirada histórica-antropológica a las posibilidades del tiempo en el relato historiográfico. *Contextos: Estudios de humanidades y ciencias sociales*, 52: 1–21.
- [50] Nidhi Tewathia, Anant Kamath, and P. Vigneswara Ilavarasan. 2020. Social inequalities, fundamental inequities, and recurring of the digital divide: Insights from India. *Technology in Society* 61: 101251. <https://doi.org/10.1016/j.techsoc.2020.101251>
- [51] Kentaro Toyama. 2011. Technology as amplifier in international development. In *Proceedings of the 2011 iConference (iConference' 11)*, 75–82. <https://doi.org/10.1145/1940761.1940772>
- [52] Martin Weller. 2015. MOOCs and the Silicon Valley Narrative. *Journal of Interactive Media in Education* 2015, 1. <https://doi.org/10.5334/jime.am>
- [53] Ben Williamson. 2021. Making markets through digital platforms: Pearson, edubusiness, and the (e)valuation of higher education. *Critical Studies in Education* 62, 1: 50–66. <https://doi.org/10.1080/17508487.2020.1737556>